

FORWARD BUILDING SYSTEM

Barricade® Thermo-Brace® SIB™ Feasibility Analysis

Application: Forward Home Kits | Forward Shelter Kits | Hybrid CLT/4Ward-SIP System

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1. Executive Summary

This analysis evaluates the Barricade® Thermo-Brace® SIB™ (Structural Insulated Sheathing) as a face material and thermal layer for the Merge4Ward Forward Building System product line, specifically across three applications:

- **Forward Home Kits** — standard residential production units
- **Forward Shelter Kits** — rapid-deployment, climate-adaptive, disaster-grade units
- **Hybrid CLT/4Ward-SIP System** — Merge4Ward’s proprietary engineered panel combining CLT structural spine with custom SIP panels

The Thermo-Brace SIB is a 4-in-1 product: structural sheathing + air barrier + water-resistant barrier (WRB) + continuous insulation (R-3 or R-5 via BASF Neopor® GPS foam). It is manufactured by Barricade Building Products (INDEVCO North America, Doswell, VA) and is TER-evaluated under reports 1507-07, 1507-08, 1507-09 for IRC R703.2 and IBC 1404.2 compliance.

Overall feasibility verdict: HIGH for Forward Home Kits. CONDITIONAL for Shelter Kits and Hybrid CLT/SIP, pending two specific confirmations. The product is a material fit for Merge4Ward’s panel manufacturing intent and directly supports the Forward Housing System’s cost compression, labor reduction, and climate-resilience design goals.

2. Product Overview: Thermo-Brace SIB vs. Thermo-Brace (Base)

Barricade offers two relevant structural sheathing products. Understanding the distinction is critical for Merge4Ward’s manufacturing decision:

2A. Thermo-Brace® (Base) — 3-in-1

A lightweight fiber-based structural sheathing panel combining: structural shear resistance, weather-resistant barrier (WRB), and air-infiltration barrier (AIB). It replaces OSB + house wrap as a 3-in-1 system. No insulation value. Available in Green (0.075”), Red (0.095”), and Blue (0.120”) grades.

- **Structural:** 16” OC (Red/Green) or 24” OC (Blue)
- **Weight:** Less than half the weight of OSB (275–350 lbs/MSF vs. ~50 lbs/sheet for OSB)
- **WRB code path:** IRC R703.2 / IBC 1404.2 — TER-evaluated
- **Custom sizing:** Up to 12’ length; up to 60” wide; no upcharges for longer lengths
- **UV tolerance:** 30-year limited system warranty; 1-year UV exposure for faced panel

2B. Thermo-Brace® SIB™ — 4-in-1 (Primary Focus)

The SIB upgrades the base product by laminating BASF Neopor® GPS (graphite polystyrene) foam to the Thermo-Brace face, adding continuous insulation. It is “reversible” — the foam face can be installed inward or outward, offering installation flexibility in the field or factory.

- **R-3 version:** TER 1709-07; foam outward uses 15/16” × 1-3/4” leg 16 ga. staples
- **R-5 version:** TER 1703-16; foam outward uses 15/16” × 2” leg 16 ga. staples; ring shank nail for inward
- **WRB on both faces:** No WRB redundancy gap regardless of foam orientation
- **GREENGUARD Gold certified:** Neopor GPS; indoor air quality confirmed
- **Energy code:** 2012, 2015, 2018, and 2021 IECC compliant; R-5 SIB satisfies northern climate CI requirement
- **UV (unfaced foam):** 90-day limit — site storage protocol required
- **Standard sizes:** 48” × 96” through 48-3/4” × 144”; custom available

STRATEGIC NOTE: The SIB’s reversible orientation is a material advantage for Merge4Ward panel manufacturing. Factory conditioning (foam inward for structural connection) and field install (foam outward for CI credit) can be spec’d independently based on panel position in the wall assembly.

3. Material Comparison: SIB vs. OSB vs. Traditional SIP

The table below benchmarks Thermo-Brace SIB against the two incumbent materials in Merge4Ward’s design space:

Attribute	Thermo-Brace SIB	Traditional OSB (SIP face)	Standard SIP Panel
Structural Rating	Up to 24" OC (Blue) / 16" OC (Red/Green)	Up to 24" OC (OSB typical)	Up to 24" OC (OSB panels)
Weight per sheet	~275–350 lbs/MSF (very light)	~50–60 lbs/sheet (heavy)	~50–60 lbs/sheet (heavy)
Built-in WRB	YES – both faces	NO – separate wrap required	YES – typically foil or poly facer
Built-in Air Barrier	YES – meets AIB	NO	YES
Insulation (R-value)	R-3 or R-5 (GPS foam)	None (R-0.62 OSB face)	Varies: R-5 to R-40+ (EPS/Poly core)
Reversible	YES – foam in or out	NO	NO – fixed orientation
Shear Value	Meets equiv. or better than OSB	High – OSB baseline	High – depends on face spec
UV Exposure	1 year (panel); 90 days (unfaced foam)	With coating only	Limited – site storage risk
Code Compliance	2012–2021 IECC; TER 1507-07/08/09	IBC/IRC all editions	ICC ESR varies by mfr
Warranty	30-year limited system	Varies; typically 1–5yr	Varies; 10–25yr typical

Custom sizing	Up to 144" length	Up to 192" typical	Up to 240" typical SIP panel
Pricing stability	Stated as stable vs. OSB market	Volatile (lumber index)	More stable; foam price driver
Fire resistance	Meets AIB; GPS foam GREENGUARD Gold	Requires separate layer	EPS/Poly core requires fire blocking

4. Application-Level Feasibility Analysis

4A. Forward Home Kits — Verdict: HIGH

The Forward Home Kit is a factory-assembled residential panel system intended for mid-cycle production at the Merge4Ward facility. Thermo-Brace SIB directly addresses the two primary cost drivers in kit production: material layer count and labor time.

Advantages in this application:

- Eliminates separate house wrap step — WRB is integrated on both faces of the SIB
- Eliminates separate continuous insulation layer for Code Zone 4 (Delaware / Philadelphia region) at R-5 SIB
- Reversible installation accommodates both factory pre-assembly and field finishing sequences
- Stable pricing vs. OSB market volatility reduces pro-forma risk in M4Di OI Tool projections
- Custom logo printing available from Barricade — co-branding opportunity for Merge4Ward product identity
- 30-year system warranty is defensible in forward housing financing and CDFI grant applications

Constraints to manage:

- Adhesive bond between Neopor GPS foam and Thermo-Brace plies in laminated panel manufacturing context — Barricade's TER reports cover field installation, not factory composite panel production. Requires direct supplier confirmation.
- Seam tape (Barricade 3" Construction Seam Tape) must be specified in kit panel schedule as a system component

4B. Forward Shelter Kits — Verdict: CONDITIONAL

The Forward Shelter Kit targets disaster-grade, rapid-deployment scenarios including post-El Niño reconstruction cycles (PHOENIX thesis). The structural and thermal performance of Thermo-Brace SIB is adequate for light wood framing applications. Specific constraints apply:

Favorable factors:

- Blue grade (0.120") rated to 24" OC — sufficient for shelter framing at standard stud spacing
- Lightweight material structure (less than half the weight of OSB) — critical for field logistics in disaster deployment scenarios
- WRB on both faces provides protection during staged delivery without additional wrap
- R-3 version adequate for climate zones where shelter kits are likely deployed (warm/mixed climates)

Risk factors requiring mitigation:

- UV exposure limit on unfaced foam: 90 days. In forward-staging or warehouse scenarios exceeding 3 months, faced-side-out protocol must be enforced. Storage specification required in kit documentation.
- High-wind zones (hurricane-prone coastal areas, post-ENSO reconstruction markets): Confirm field engineering report (FER) applicability under local wind load criteria. TER evaluation is IRC/IBC-based; some jurisdictions require supplemental wind testing.
- Fire blocking at floor and ceiling lines must be detailed — foam core creates continuity gap not present in OSB SIPs. Same requirement as standard EPS SIP construction.

RISK MANAGEMENT: 90-day unfaced UV limit is a field logistics constraint, not a product disqualifier. Manage with 'foam-in' staging protocol for stored kits exceeding 60 days. Flag in Shelter Kit specification sheet.

4C. Hybrid CLT/4Ward-SIP System — Verdict: **CONDITIONAL**

The Hybrid CLT/4Ward-SIP System — as demonstrated in the Claymont project (AE24-002) and the Merge4Ward engineering intent — uses CLT as the primary structural spine with SIP panels for non-load-bearing wall and roof enclosure. Thermo-Brace SIB is evaluated here as the face material for the 4Ward-SIP component, not as a replacement for the CLT structure.

Structural role clarification:

- In a CLT-spine hybrid, the SIP face material carries lateral racking loads within the panel only — primary vertical and horizontal structural loads are resolved by CLT
- Thermo-Brace SIB's structural rating (equivalent to or better than OSB per TER) is adequate for this supplemental panel role
- The composite action between SIB face, foam core (Neopor GPS), and CLT spine must be engineered by EOR — no TER covers this specific assembly

Key unknowns requiring EOR engagement:

- Adhesive compatibility between Neopor GPS and CLT edge connections (sealant and mechanical connector interface)
- Party wall fire rating when SIP panel abuts CMU party wall (AE24-002 scenario): Thermo-Brace SIB does not independently satisfy 2-hour rating; CMU continuity is the confirmed fire wall strategy
- SIB panel connection to CLT at bearing points — bottom plate and anchor bolt requirements per ESR-4689 analogy must be confirmed for non-CLT-bearing SIB panels

CRITICAL: Do NOT spec Thermo-Brace SIB as a drop-in face for an ESR-4689 compliant SIP system without EOR review. The TER covers the SIB as sheathing/WRB; a full composite SIP assembly using SIB faces requires separate structural engineering. This is an opportunity — Merge4Ward can own the proprietary ESR for the 4Ward-SIP assembly.

4D. Merge4Ward Manufacturing — Verdict: FEASIBLE WITH CAVEATS

From a manufacturing integration standpoint, Thermo-Brace SIB is compatible with Merge4Ward's intended panel production workflow at the CLT/SIP facility. Specific manufacturing advantages and constraints:

Manufacturing advantages:

- Single-source panel face eliminates OSB procurement and separate WRB lamination step
- Barricade's custom sizing (up to 144" length) reduces field cutting waste and aligns with factory panel module planning
- INDEVCO North America Doswell, VA facility — Mid-Atlantic supply chain, close to Wilmington; logistics favorable
- Stable pricing vs. OSB market allows more reliable unit pricing in ALDDB; supports M4Di OI Tool ROM accuracy
- Custom logo printing available — Merge4Ward brand on every panel shipped is a product identity asset

Manufacturing constraints:

- Factory lamination of Thermo-Brace SIB to foam core in Merge4Ward's own panel production requires adhesive bond confirmation from Barricade. The TER-evaluated assembly is field-installed SIB; factory composite manufacturing is a different production context.
- Neopor GPS is BASF's proprietary EPS formulation. If Merge4Ward sources foam separately for its 4Ward-SIP panels, compatibility with Thermo-Brace plies must be tested. If SIB (which comes pre-laminated) is the product, this is moot — the panel is the product.
- Fastener schedule differences between R-3 and R-5 SIB must be reflected in panel shop drawings

5. Cost and Pricing Implications

Pricing for Thermo-Brace SIB is not publicly listed; quoting is by direct request. Based on available market context:

- Barricade states stable pricing compared to OSB market volatility — material advantage for M4Di ROM estimation and ALDDB unit price stability
- The 4-in-1 system (SIB = sheathing + WRB + AIB + CI) eliminates three separate line items in a traditional wall assembly. Net cost impact is favorable even if SIB carries a per-sheet premium over bare OSB
- Labor reduction is the primary cost driver: one install step vs. four (sheathing, tape, wrap, CI layer). For Merge4Ward factory production, this compresses panel assembly time materially
- Custom sizes reduce field cut waste — relevant for M4Di kit costing in ALDDB (unit price should reflect net installed vs. gross ordered)

ACTION: Merge4Ward should obtain a volume pricing quote from Barricade (INDEVCO NA Doswell) in Q3 2026 concurrent with Thermo-Brace SIB panel spec confirmation. Lock a unit price for ALDDB entry and M4Di OI Tool ROM calibration before Q4 2026 engineering spec target.

6. Code Compliance Summary — Mid-Atlantic Region

Thermo-Brace SIB has the following confirmed code compliance pathways relevant to Merge4Ward's primary market (Delaware, Pennsylvania, Maryland; City of Wilmington Ch. 48; NCC UDC):

- **WRB:** IRC Section R703.2 and IBC Section 1404.2 — TER evaluated. Satisfies water-resistive barrier requirement as sole WRB layer when taped at seams.
- **Structural sheathing:** Evaluated as equivalent to or better than OSB for racking resistance. Alternative corner bracing method accepted per TER. Submit TER with permit documents.
- **Energy code:** 2012, 2015, 2018, 2021 IECC compliant. R-5 SIB satisfies continuous insulation (CI) requirement for Climate Zone 4 (Delaware, southeastern PA, Maryland). Eliminates need for separate CI layer on forward-facing exterior wall.
- **Air barrier:** Meets AIB requirements. Integrated with WRB face — reduces AHJ inspection steps.
- **Indoor air quality:** Neopor GPS is GREENGUARD Gold certified. Satisfies IAQ documentation requirements for CDFI and HUD financing sustainability standards.

COMPLIANCE NOTE: City of Wilmington uses Ch. 48 Zoning Ordinance and adopts IBC/IRC with Delaware amendments. TER reports 1507-07/08/09 are the AHJ submission documents for structural WRB equivalency. Include TER in permit package on all Merge4Ward forward housing projects.

7. Application Verdict Summary

Application	Verdict	Rationale
Forward Home Kit (stick-frame SIP hybrid)	HIGH	Thermo-Brace SIB replaces separate WRB, insulation board, and OSB in kit-panel assembly. Significant labor and material compression. Reversible install suits field variation.
Forward Shelter Kit (rapid deployment, disaster-grade)	CONDITIONAL	Feasible for wall panels. Structural rating at 24" OC sufficient for light framing. Evaluate foam durability in prolonged UV exposure (90-day unfaced limit is a constraint for staging/storage). Confirm FER for hurricane/high-wind zones.
Hybrid CLT/4Ward-SIP System (Claymont-type / Merge4Ward product)	CONDITIONAL	Thermo-Brace SIB is a viable outer face for Merge4Ward-manufactured SIP panels, replacing OSB. Foam core bond compatibility with EPS/Neopor GPS must be tested per manufacturer. Structural adequacy depends on composite panel engineering (CLT spine + SIB face — requires EOR sign-off). Not a drop-in replacement for full structural SIP; supplement role only.
Merge4Ward Manufacturing (4Ward-SIP product line face material)	FEASIBLE WITH CAVEATS	Can be incorporated as the structural face layer in lieu of OSB. Eliminates separate WRB step in panel production. Factory-applied seam tape integrates with existing Barricade system. Key constraint: adhesive bond between Neopor GPS foam and Thermo-Brace plies needs supplier confirmation for laminated panel manufacturing (not just field install).
Code Compliance (Mid-Atlantic: DE, PA, MD)	CONFIRMED	TER reports 1507-07, 1507-08, 1507-09 satisfy IRC R703.2 and IBC 1404.2 as WRB. IECC 2021 compliance confirmed (R-5 northern zone). Compatible with City of Wilmington and NCC building code requirements.

8. Key Risks and Mitigations

Risk 1: Factory Lamination Bond Incompatibility

The TER-evaluated SIB assembly is field-installed. If Merge4Ward produces composite panels in-factory using SIB faces, adhesive bond integrity under production pressure/heat cycling is unconfirmed. **Mitigation:** Direct supplier technical call with Barricade/INDEVCO; request factory lamination suitability letter.

Risk 2: UV Degradation in Storage

Unfaced foam faces are rated at 90 days UV exposure. Extended site or warehouse storage for Shelter Kit staging could degrade the foam. **Mitigation:** Enforce foam-inward staging protocol for all stored panels. Flag 60-day review trigger in kit logistics documentation.

Risk 3: High-Wind Zone Structural Gap

TER structural ratings are IRC/IBC-based. Some coastal or hurricane-zone AHJs (post-ENSO reconstruction markets) may require supplemental wind-resistance testing. **Mitigation:** Confirm FER applicability in target deployment markets before Shelter Kit spec is locked. Contact Barricade for coastal wind test data.

Risk 4: Fire Resistance at Party Walls and Floor/Ceiling Lines

GPS foam core, like all EPS, requires fire blocking at floor and ceiling lines per IRC R302.12. Thermo-Brace SIB does not independently provide 2-hour fire wall rating. **Mitigation:** Maintain CMU continuous party wall strategy (confirmed for AE24-002). Specify fire blocking in panel schedule on all permitted projects.

Risk 5: Pricing Opacity

No public list pricing. ROM estimates for ALDDB and M4Di OI Tool require negotiated volume quotes. **Mitigation:** Obtain Q3 2026 volume quote and lock unit price in ALDDB. Barricade's stated pricing stability supports multi-year unit price hold.

9. Recommended Next Steps

Action	Owner	Target	Why It Matters
Contact Barricade rep — adhesive bond compatibility for factory lamination of SIB to foam core	David / Travis	Q3 2026	Validates factory use vs. field-only install
Request TER 1709-07 (SIB structural evaluation report) from Dr. J Certification	David	Q3 2026	Confirms structural rating path under IRC/IBC for AHJ submissions

Specify SIB in Forward Home Kit panel schedule; submit for EOR review	David / EOR	Q3–Q4 2026	Locks spec before engineering target (Q4 2026)
Evaluate R-5 SIB against IECC CZ4 (Delaware) — confirm continuous insulation credit	David	Q3 2026	Confirms energy code compliance without separate CI layer
Pilot installation on Merge4Ward prototype panel — document shear testing per TER	David / Trevor / Travis	Q4 2026	First real-world proof for ALDDB unit price entry
Engage Barricade on custom logo/co-brand printing option for Merge4Ward panels	David / Trevor	Q4 2026	Brand/IP moat; Barricade offers this as standard option

10. Conclusion

Thermo-Brace SIB is the most compelling single-product substitution available to Merge4Ward for the Forward Building System product line. It eliminates OSB, house wrap, and a separate CI layer in a single panel — three of the five most time-intensive assembly steps in a conventional wall assembly. The labor compression, pricing stability, Mid-Atlantic supply chain, 30-year warranty, and IECC 2021 compliance make it a strong candidate for specification across all three Forward Housing product types.

Two confirmations are required before the specification is locked for manufacturing: (1) factory lamination bond suitability from Barricade/INDEVCO, and (2) EOR review of SIB use in the Hybrid CLT/4Ward-SIP composite assembly. Neither is a blocking issue — both are resolvable in Q3 2026 with targeted supplier and engineering outreach.

Merge4Ward’s co-branding opportunity (custom logo printing on every SIB panel) is a secondary but meaningful IP signal — the product ships with Merge4Ward identity. Combined with the ALDDB unit price entry and M4Di OI Tool ROM calibration, this positions Thermo-Brace SIB as a foundational component of the Forward Housing System’s data moat and product differentiation strategy.

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Source documents: Barricade Thermo-Brace SIB product page; Barricade Brochure & Specifications (Jan 2023); Barricade Installation Guide (2023); TER 1507-07, TER 1507-08, TER 1507-09, TER 1709-07; 30-Year System Warranty (Jan 2023); GreenBuildingAdvisor fiberboard analysis; Capital Forest Products engineered panel comparison (2024).

11. Alternative Materials Considered

Before confirming the Forward Building System panel material specification, nine structural sheathing and enclosure panel candidates were evaluated across eight criteria: fire resistance, structural rating, built-in WRB, continuous insulation integration, US supply chain availability, ICC-ES or equivalent code acceptance, kit/panel manufacturing fit, and overall strategic alignment with Merge4Ward’s product

architecture and brand narrative. The evaluation synthesized product technical data, ICC-ES evaluation reports, TER documentation, and market research current as of May 2026.

11A. Overview comparison matrix

The table below summarizes all nine candidates. Ratings reflect confirmed technical data and code acceptance paths, not manufacturer claims alone.

Material / Manufacturer	Class	Fire rated	Structural	WRB built-in	CI insulation	US supply chain	ICC-ES / TER	Merge4Ward verdict
Thermo-Brace SIB (Barricade / INDEVCO)	Fiberboard + EPS	Class C	YES - 24" OC	YES - both faces	R-3 / R-5	YES - VA	YES - DrJ TER	CURRENT LEAD - standard enclosure
Huber EXACOR (Huber Engineered Woods)	MGO (oxysulfate)	1-2 hr ASTM E119	YES	Separate WRB	None	YES - US domestic	YES - ESR-4635	TOP ALTERNATIVE - fire-rated panel face
USG Structo-Crete (USG / Knauf)	Cementitious concrete	1-3 hr ASTM E119	YES	Separate WRB	None	YES - Delavan WI	YES - ESR-1792	Strong but heavy - subfloor / roof deck priority
MagMatrix BMSC 517 (Jinpeng Group)	MGO sulfate (BMSC)	1-2 hr ASTM E119	YES	Separate WRB	SIS version only	Import only	ASTM only - no ICC-ES ESR confirmed*	Best chemistry - import + ESR gap to resolve first
LP FlameBlock (LP Building Solutions)	Coated OSB	Flame-block rated	YES	Separate WRB	None	YES - US wide	YES - ESR-1365	Good budget option; OSB core limits brand narrative
AAC panels (Xella / import mfrs)	Autoclaved aerated concrete	Up to 4 hr	Infill only	Separate WRB	Inherent R-value	Limited US supply	ASTM C1386 only	Wrong form factor for kit panels - too thick / heavy
Fiber cement sheathing (James Hardie / Nichiha)	Fiber cement	Non-combustible	Limited shear	Partial only	None	YES - US wide	Multiple ESRs	Cladding / siding role - wrong layer for panel face
Atlas ThermalStar SWI (Atlas Molded Products)	EPS + film facer	None	Limited	YES	R-3 / R-5	YES - US domestic	Partial	Closest competitor to Thermo-Brace SIB
ZIP System R-sheathing (Huber Engineered Woods)	OSB + foam + tape	None	YES	YES - tape system	R-6 to R-12	YES - US wide	YES - full ESR	Strong system; OSB core - not non-combustible

* BMSC 517 holds ASTM E119/E136/E84 test reports and EN 13501-1 A1/A2 certifications. No confirmed active ICC-ES ESR as of May 2026. Verify current status at icc-es.org before AHJ submission. The DrJ TER pathway (used by Thermo-Brace) is an alternative if Barricade / INDEVCO pursues it. | Scoring key: YES = confirmed / strong. Partial / conditional = requires additional steps. None / absent = not applicable.

11B. Candidate-by-candidate findings

Thermo-Brace SIB (Barricade / INDEVCO) — Confirmed current lead for standard enclosure panels. 4-in-1 system (structural + WRB + AIB + R-3/R-5 CI) eliminates OSB, house wrap, and CI layer in one factory or field step. TER 1507-09 provides full structural shear values, seismic design coefficients ($R=6.5$, $C_d=4$), uplift resistance (400 plf), and transverse wind resistance (200 mph Vult basic wind speed). See Sections 2-4 of this report for full analysis.

Huber EXACOR (Huber Engineered Woods, ESR-4635) — Top alternative identified for fire-rated panel face applications. MGO oxysulfate chemistry, no corrosion risk to wood or steel framing, 1- and 2-hour ASTM E119 fire-rated wall assemblies confirmed for Types III and V construction. Active ICC-ES ESR-4635 and listing ESL-1290. Made in the US. Compatible with wood framing including CLT-spine hybrid assemblies. The confirmed code path and domestic supply chain make this the preferred specification for the 4Ward-Fire Panel role over BMSC 517 in the near term. EXACOR replaces BMSC 517 as the primary fire panel face specification pending BMSC 517 obtaining a US ICC-ES ESR.

USG Structo-Crete (USG / Knauf, ESR-1792) — Only noncombustible cementitious structural panel manufactured in the US (Delavan, WI). ASTM E136 compliant for all construction types; 1-, 2- and 3-hour UL fire assemblies. Shear capacity exceeds OSB; STC 50-70 range. Primary constraint for kit manufacturing is panel weight (approximately 3,400 lbs per 30-sheet pallet of half-inch panels), which creates field logistics friction for a forward housing deployment model. Best fit: subfloor and roof deck applications where weight is less problematic, not exterior wall panel faces for kit systems.

MagMatrix BMSC 517 (Jinpeng Group) — Technically superior chloride-free sulfate MGO chemistry that eliminates the halogenation / fastener corrosion failure mode of first-generation MGO. Holds ASTM E119, E136, and E84 test reports and EN 13501-1 A1/A2 certifications. A Structural Insulated Sheathing (SIS) version combines fire rating with continuous insulation in a single panel — potentially the most differentiated future product in this space. Current constraint: no confirmed active ICC-ES ESR as of May 2026, and supply chain is import-only from China. Specification as primary fire panel face is deferred until ICC-ES or DrJ TER code acceptance is confirmed. Retained as a watch-and-activate candidate: if and when the ESR gap is resolved, BMSC 517 SIS replaces EXACOR as the fire panel specification due to its combined fire + CI capability.

LP FlameBlock (LP Building Solutions, ESR-1365) — APA-certified OSB with noncombustible Pyrotite cementitious coating. Full ICC-ES evaluation, UL and Intertek listing, national US distribution. The most cost-accessible fire-rated sheathing option and familiar to every framing crew. Primary limitation: OSB substrate is combustible — the panel is a fire-resistive system, not a non-combustible panel. Does not advance the Forward Building System brand narrative. Appropriate as a budget-tier option for non-fire-rated wall positions where Thermo-Brace SIB pricing is a barrier, subject to a full system code review.

AAC panels (Xella / import manufacturers) — Autoclaved aerated concrete panels offer exceptional fire performance (up to 4-hour ratings), inherent thermal mass, mold resistance, and non-combustibility. Thermal conductivity of approximately $0.16 \text{ W/m}\cdot\text{K}$ and STC ratings of 44 dB are genuinely superior. The disqualifying factor for Merge4Ward's Forward Home Kit and Shelter Kit applications is form factor: AAC panels are 100–200mm thick, mass-based wall assemblies, not face panels for a factory-assembled SIP kit system. US supply chain is limited and the ASTM C1386 code path does not include structural shear wall equivalency. AAC remains viable as a party wall block strategy in conventionally framed ground-story construction — consistent with the CMU approach confirmed for AE24-002 — but is not a panel face candidate.

Fiber cement sheathing (James Hardie / Nichiha) — Non-combustible, moisture-resistant, dimensionally stable, US domestic supply, multiple ICC-ES code paths. Strong material in the right role. The limitation here is that fiber cement is a cladding and siding substrate, not a structural shear panel face. Shear values are limited, panels are brittle in handling, and they require a separate WRB layer. Appropriate for exterior finish systems on top of the 4Ward-SIB or 4Ward-Fire panel, not as the panel face itself.

Atlas ThermalStar SWI (Atlas Molded Products) — The closest direct competitor to Thermo-Brace SIB in the market: EPS laminated to a structural film facer, R-3 and R-5 options, WRB integrated, US domestic supply. Reversible installation is not confirmed as a feature. Code acceptance path is partial — check current ICC-ES listing status before specifying. Should be used as a price benchmark against Thermo-Brace SIB when negotiating volume pricing with Barricade/INDEVCO.

ZIP System R-sheathing (Huber Engineered Woods) — The market-leading integrated sheathing + WRB + CI system with higher R-value options (R-6 to R-12) than Thermo-Brace SIB and a full structural rating backed by a comprehensive ICC-ES ESR. Tape-integrated WRB system is field-proven. The disqualifying characteristic for Forward Building System is the OSB substrate: ZIP System is not a non-combustible panel and does not support the forward housing climate-resilience brand narrative or fire-rated party wall applications. Not a primary candidate for Merge4Ward panel manufacturing.

11C. Product architecture decision

Based on this review, the two-panel product line architecture is confirmed with one material substitution from the draft specification:

- **4Ward-SIB Panel** — standard enclosure, Forward Home Kit and Shelter Kit applications. Face material: Thermo-Brace SIB (R-3 or R-5 per climate zone). No change from draft specification.
- **4Ward-Fire Panel** — fire-rated condition: party walls, demising walls, egress corridors, CLT hybrid fire separation assemblies. Face material: **Huber EXACOR (ESR-4635)**, replacing MagMatrix BMSC 517 as primary specification. EXACOR delivers the same non-combustible MGO chemistry with an active ICC-ES ESR and US domestic supply — both required for AHJ submission in Mid-Atlantic markets today.
- **BMSC 517 watch-and-activate** — MagMatrix BMSC 517 is retained as the priority upgrade path for the 4Ward-Fire Panel if and when ICC-ES or DrJ TER code acceptance is confirmed. The BMSC 517 SIS (Structural Insulated Sheathing) version — combining fire rating, structural capacity, and continuous insulation in one panel — is the most technically differentiated product in this evaluation and would, if code-accepted, consolidate the two-panel system into a single fire-rated + insulated panel for most applications.

All fire-rated assemblies referencing EXACOR must be designed by the EOR using tested assembly configurations from ESR-4635 and ESL-1290. Merge4Ward's proprietary 4Ward-Fire Panel assembly configuration will require a separate engineering evaluation and, ultimately, a Merge4Ward-specific ESR or TER as part of the platform's intellectual property development roadmap.